ENGR 225: Mechanics 1

Homework Coversheet

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DATE: 6 November 2026

ASSIGNMENT: Design Project, Part 1

Level of involvement of AI, LLMs, or related tools in the completion of this assignment:

☐ None

☐ Minor

■ Major

☐ Extensive

Mark the box that applies (e.g., replace ☐ with ■, or type an X beside it, before compiling/printing). Your disclosure does not affect your grade – it keeps you accountable for the work you are doing.

ENGR 225 Design Project, Part 1: Roadway Selection

Sean Balbale Assigned: 26 Oct 2026 Due: 6 Nov 2026

1 Objective

The objective of Part 1 is to select and quantify the roadway that the truss bridge will carry. The specific goals are:

- To classify the roadway and select lane, shoulder, and barrier widths from AASHTO and CTDOT standards.
- To specify the deck section (slab, waterproofing, wearing surface) and calculate material quantities for the 60 m span.
- To estimate the cradle-to-gate embodied carbon of those materials and compare it against lower-carbon alternatives.
- To estimate the raw material cost.

2 Design Basis

Given Parameters

- **Span:** $L = 60$ m, two lanes per direction, no pedestrian traffic.
- **Reference roadway:** Route 2 over Griswold St., Glastonbury: 74,400 vehicles/day, 6% trucks (2021) [1], or about 4,460 trucks/day.
- **Classification:** urban principal arterial freeway (divided, full access control).
- **Standards:** AASHTO freeway criteria for widths [2]; CTDOT Bridge Design Manual and pavement guidance for the deck section [3, 4].

3 Roadway Design

Cross-Section AASHTO sets a 12 ft minimum lane width and a 10 ft minimum shoulder, except that the left shoulder of a 4-lane freeway can be 4 ft [2]. I used the minimums, since every foot of width carries into the slab, rebar, and overlay quantities. The bridge carries the full approach width. Each edge has a 32 in concrete parapet and the median has a 42 in barrier; the barrier base widths (21 in and 24 in) are my assumptions, not CTDOT plate values.

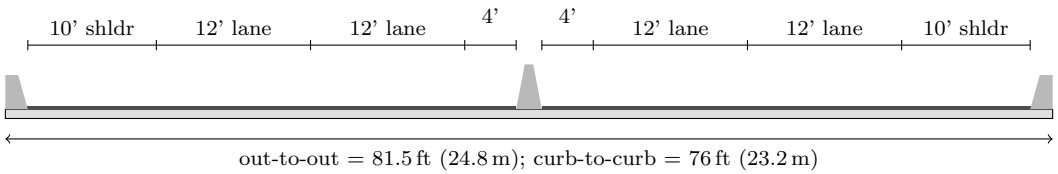


Figure 1: Roadway cross-section: 3 in HMA on membrane over an 8.5 in reinforced concrete slab (vertical not to scale).

Deck Section

- **Slab:** 8.5 in (216 mm) cast-in-place reinforced concrete, the CTDOT minimum, with 2.5 in cover over the top mat [3]. Reinforcement is assumed at 150 kg/m^3 until Part 2 sizes it.
- **Waterproofing:** spray-applied liquid membrane, CTDOT's current standard under a bituminous surface [4].
- **Wearing surface:** 3 in hot-mix asphalt (HMA): a 1 in S0.25 lift on the membrane and a 2 in top course [4].

Quantities With out-to-out width $W = 24.84\,\text{m}$ and curb-to-curb width $w_c = 23.16\,\text{m}$:

$$
\begin{aligned}
V_{\text{deck}} &= LWt_s = (60\,\text{m})(24.84\,\text{m})(0.216\,\text{m}) = 322\,\text{m}^3 \\
V_{\text{barrier}} &= L(2A_p + A_m) = (60\,\text{m})(2 \times 0.279\,\text{m}^2 + 0.434\,\text{m}^2) = 59.5\,\text{m}^3 \\
m_{\text{rebar}} &= (381\,\text{m}^3)(150\,\text{kg/m}^3) = 57.2\,\text{t} \\
m_{\text{HMA}} &= Lw_c t_o \rho_{\text{HMA}} = (60\,\text{m})(23.16\,\text{m})(0.076{,}2\,\text{m})(2.35\,\text{t/m}^3) = 249\,\text{t}
\end{aligned}
$$

Total concrete is 381 m^3 (499 yd^3).

4 Environmental Impact

Embodied Carbon I used environmental product declarations (EPDs), which report cradle-to-gate (A1–A3) CO_2e per unit of material. A Connecticut-area 4,000 psi EPD covers mixes from 265 kg/m^3 to 335 kg/m^3 [5], so I used the 300 kg/m^3 midpoint. Rebar is 854 kg/t from the CRSI industry average [6], and HMA is 71.1 kg/t from an FHWA example EPD [7].

Table 1: Baseline embodied carbon (A1–A3)

Material	**Quantity**	**GWP Factor**	**CO_2e (t)**
Concrete (deck + barriers)	381 m^3	300 kg/m^3	114
Reinforcing steel	57.2 t	854 kg/t	48.8
HMA overlay	249 t	71.1 kg/t	17.7
Total			**181**

Concrete is 63% of the total, and nearly all of that is the portland cement, which emits CO_2 from both kiln fuel and limestone calcination. The membrane is excluded since I could not find an EPD for it; it is a few millimeters thick, so its mass is small next to the slab.

Table 2: Embodied carbon of alternative materials (same geometry)

Option	CO_2e (t)	Change
Baseline	181	–
Slag-cement concrete, 40–50% lower GWP [8]	124–135	−25 to −32%
EAF rebar on renewable power, 300 kg/t [9]	149	−18%
Concrete wearing surface instead of HMA	195	+8%
Slag concrete (45%) + EAF rebar	98	−46%

Alternatives Cement content is the biggest lever: a 60% slag mix approved for Australian infrastructure cut embodied carbon by an estimated 40–50% [8], at the cost of slower early strength gain. HMA is also the better wearing surface on carbon, not just by CTDOT practice; replacing it with 106 m^3 of concrete adds 14 t.

Perspective EPA puts a typical passenger vehicle at about 400 g CO_2 per mile [10], so traffic over the span emits

$$(74{,}400/\text{day})\left(\frac{60\,\text{m}}{1{,}609\,\text{m/mi}}\right)(0.400\,\text{kg/mi}) = 1.11\,\text{t/day}.$$

The baseline materials equal about 163 days of that traffic. This slightly overstates the ratio since trucks emit more per mile than cars, but the materials are the part of the footprint the designer controls.

Other Impacts The concrete and HMA contain several hundred tonnes of quarried aggregate, with the land disturbance and truck traffic that comes with it. The A1–A3 figures also stop at the plant gate, so a distant plant with a lower EPD value can come out worse once hauling is counted. Recycled asphalt pavement (RAP) in the HMA cuts virgin aggregate and binder demand.

5 Raw Material Cost

Table 3: Raw material cost (2026 USD, excluding labor and formwork)

Material	Quantity	Unit Price	Cost
Ready-mix concrete	499 yd^3	\$185/yd^3 [11]	\$92,200
Reinforcing steel	57.2 t	\$1,100/t [12]	\$62,900
HMA	274 short tons	\$130/ton [13]	\$35,700
Total			**\$191,000**

Concrete is priced at the top of the \$155–\$185/yd^3 Northeast range [11], since a deck mix is air-entrained and higher strength than residential concrete. Rebar rounds up from the \$1,058/t US average at the start of 2026 [12] to cover fabrication. HMA is a CTDOT bid price for S0.5 (\$129–\$135/ton) [13], which includes placement, so it is an upper bound. The same contract bid deck concrete at \$700–\$915/yd^3 installed, so raw concrete is only 20–26% of the in-place cost.

6 Conclusion

The roadway is a 4-lane urban freeway deck, 60 m long and 24.8 m wide out-to-out.

1. **Geometry:** four 12 ft lanes, 10 ft right and 4 ft left shoulders, 76 ft curb-to-curb. Widening the left shoulders to 10 ft would add about 15% to the deck and overlay quantities.
2. **Deck section:** 8.5 in reinforced concrete slab, liquid membrane, and 3 in HMA, giving 381 m^3 of concrete, 57.2 t of rebar, and 249 t of HMA.
3. **Embodied carbon:** 181 t CO_2e, 63% from concrete. Slag cement and EAF rebar together bring it to about 98 t (−46%).
4. **Cost:** about $191,000 in raw materials, 48% of it concrete.

Slab thickness and rebar density are the values most likely to change once Part 2 sizes the truss and the deck span between supports.

AI Use I used Claude (Anthropic) to locate the CTDOT, FHWA, and EPD sources, check the quantity and carbon arithmetic, and draft and edit this report. I checked each design value against its cited source and made the classification, shoulder-width, and material decisions myself.

References

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